Assignment 6 — Gene Co-Expression Networks (TP53) — unknown13m

1) Date si preprocesare
- log2(x+1) pe expresie
- filtrare gene cu varianta scazuta (pastrate genele cele mai informative)

2) Retea, module (Louvain)
- corelatie gene-gene (Pearson/Spearman)
- prag |corr| pentru adjacency si graf
- detectie module cu Louvain

3) Hub genes

Hub genes selectate (top dupa strength/degree):

gene	module	degree	strength
Sample6	1	6	2.426
Sample4	1	4	1.836
Sample20	2	3	1.486
Sample10	1	3	1.272
Sample16	2	3	1.217
Sample12	2	3	1.138
Sample7	2	3	1.099
Sample15	1	2	1.067
Sample14	1	2	0.924
Sample3	0	1	0.359

4) Interpretare biologica si diseaseome
- Modul ales pentru analiza: module 1
- Enrichment (GO/KEGG): rulati g:Profiler / DAVID pe genele din modul ales.
  In raportul final (manual), lipiti top 5 termeni GO/KEGG + p-values.

Reflectie (diseaseome):

Acest modul ar putea fi legat de boala daca genele sunt co-activate in acelasi context biologic (ex: raspuns la stres, reparare ADN, apoptoza). In diseaseome, hub-urile pot actiona ca puncte de convergenta intre fenotipuri si cai moleculare, sugerand subtipuri sau potentiale tinte terapeutice.